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Field-Based MIC Management and Biocide Optimisation Through Onsite Molecular Monitoring and Biocide Residual Tracking

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This paper presents an operational case study focused on maintaining asset integrity in offshore production systems by managing Microbiologically Influenced Corrosion (MIC). The objective is to share how a field-executed Corrosion Monitoring Programme (CMP), integrating molecular microbiology, process chemistry, and biocide tracking, has been used to reduce corrosion risks, optimise treatment, and enhance operational decision-making.

Microbiologically Influenced Corrosion (MIC) presents a persistent threat to asset integrity in offshore oil and gas systems. MIC is caused by the metabolic activities of microorganisms—especially sulfate-reducing bacteria (SRB), acid-producing bacteria (APB), and other anaerobes—that generate corrosive by-products and biofilms, accelerating internal corrosion rates in carbon steel infrastructure. The complexity of MIC is exacerbated in seawater and produced water systems, where nutrient availability, flow dynamics, and microbial ecology intersect.

This paper presents an operational case study spanning from 2022 to 2025, where an integrated Corrosion Monitoring Programme (CMP) was deployed in an offshore asset. The CMP comprised microbial diagnostics (ATP, AMP, qPCR), water chemistry surveillance, and process parameter tracking. This data triad provided insights that enabled field teams to shift from reactive to predictive MIC mitigation strategies, including optimised biocide dosing and timely intervention.

Introduction

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Background and Sampling Approach

An offshore gas facility in the region began to report the presence of black precipitates and H₂S readings in their seawater treatment and produced water system. The presence of the black precipitates was identified as Iron Sulfide. A project was then conducted to understand the issues and Microbiologically Influenced Corrosion (MIC) was identified as the cause.

Sampling Points and Scope

Sampling was carried out at 15 strategic points across both the seawater treatment and produced water treatment systems. These points were selected to represent various the seawater treatment and produced water high priority points for the operator. The primary sampling medium at the early stages of the programme was water, targeting planktonic (free-floating) microorganisms.

Planktonic vs Sessile Sampling

Understanding microbial colonisation in a system requires distinction between planktonic and sessile bacterial communities.

- **Planktonic bacteria** are suspended in the fluid phase. These organisms are easier to access, especially at the early stages of a monitoring programme when dedicated biofilm collection infrastructure such as biostuds or Modified Robbins Devices (MRDs) may not yet be installed. Sampling planktonic bacteria is therefore a practical and cost-effective initial step for operators launching microbial surveillance.
- **Sessile bacteria** are those adhered to surfaces, often embedded within biofilms. Sessile communities play a more direct role in MIC due to their stable colonisation and close contact with metal surfaces. Sessile sampling (e.g. via biostuds or coupons) is the preferred method for MIC diagnosis but is logistically more involved.

In this case study, the operator elected to start with planktonic monitoring, supported by process data and corrosion indicators, as a feasible first step.

Microbiological Analysis Method

ATP and AMP

Adenosine Triphosphate (ATP) is a universal energy-carrying molecule found in all living cells. Its presence and concentration are widely used as proxies for microbial biomass and activity. Adenosine Monophosphate (AMP), a breakdown product of ATP, can be measured in tandem to provide a broader indication of total cellular presence - including stressed or dying cells.

In this programme:

- Total ATP was used to represent active biomass

- AMP was used to complement ATP readings and account for bacteria under stress, senescence, or antimicrobial pressure
- Combined ATP+AMP values provided a fuller microbial profile.

These tests were conducted onsite using portable luminometers and reagents, enabling near-immediate turnaround (typically under 1 hour), which is crucial for time-sensitive decision-making in offshore operations.

Sulphate Reducing Bacteria Enumeration via qPCR

Quantitative Polymerase Chain Reaction (qPCR) was employed to quantify SRB populations based on DNA analysis. Unlike culture-based methods, qPCR targets species-specific or functional gene sequences. Key features of qPCR include:

- High specificity
- Quantitative output
- Sensitivity to low bacterial loads
- Rapid turnaround (typically within 3 hours)

qPCR's ability to detect viable but non-culturable (VBNC) bacteria offers a significant advantage in MIC scenarios where stress-tolerant SRB may evade traditional detection methods.

Advantages of Molecular Microbiology Method

qPCR vs Culture-Based Methods

Culture-based enumeration, such as Most Probable Number (MPN) techniques, has long been used in oilfield microbiology. These methods rely on growing microorganisms in selective media and incubated for a period of time under process temperature. While this approach provides functional confirmation, it is inherently limited by:

- Slow turnaround times (7-28 days)
- Bias towards fast-growing strains
- Failure to detect VBNC organisms
- Qualitative or semi-quantitative output - a statistical approach

In contrast, qPCR offers rapid, accurate, and quantitative detection of SRB. While culture methods are still valuable—especially when evaluating metabolic activity or confirming viable threats—qPCR serves as a more responsive tool for field operators needing to make timely biocide and intervention decisions.

qPCR vs Rapid Test Kits

Rapid colorimetric or dipstick-based microbial test kits are commonly used for quick assessments in the field. While they offer simplicity and speed, their limitations include:

- Poor specificity (may detect unrelated microbial groups)
- High rates of false positives/negatives
- Subjective interpretation (color-based readings)

- Limited quantification

These kits can serve as preliminary indicators but are not reliable for trending, diagnosis, or treatment evaluation. In contrast, qPCR provides traceable, reproducible, and actionable data aligned with engineering and corrosion control requirements.

Role of Water Chemistry and Process Data

Understanding MIC requires more than just microbiology. Corrosion is a multi-factorial process, and the integration of water chemistry and process conditions is essential for contextual analysis.

Water Chemistry

Routine analysis included parameters such as:

- **Dissolved Oxygen (DO):** Oxygen presence fuels aerobic corrosion and influences microbial populations
- **pH and Bicarbonate:** Affect metal passivation and microbial activity
- **Volatile Fatty Acids (VFAs):** Metabolic by-products of acid-producing bacteria, which can drive localised corrosion
- **Sulfate:** A nutrient for SRB; high concentrations can support aggressive growth
- **Salinity and Total Dissolved Solids (TDS):** Impact microbial ecology and corrosion dynamics

These parameters helped interpret microbial behaviour and supported the prediction of MIC hotspots.

Process Parameters

Process data included:

- Temperature and Pressure: Affect microbial growth rates and biocide efficacy.
- Flow rates

By aligning microbial trends with process anomalies (e.g., pressure drops, temperature fluctuations), operators could better diagnose whether changes were due to microbial growth or other system variable.

Mitigation and Treatment Strategy

MIC indicators such as Total Suspended Solids (TSS) and Hydrogen Sulfide (H₂S) in process gas were tracked to monitor corrosion risks. These data streams, when overlaid with microbial and chemistry profiles, helped build a robust picture of asset condition. Treatment strategies were developed based on:

- **Molecular monitoring feedback:** ATP/AMP and qPCR trends guided chemical intervention decisions
- **Biocide rotation protocols:** THPS and Glutaraldehyde were alternated to mitigate microbial resistance
- **Residual biocide tracking:** Ensured sufficient biocide concentration in the system post-dosing

To assist decision-making, a traffic light system was implemented, categorising sites and time points by microbial risk and treatment efficacy.

Results and Discussion

Sulfate-Reducing Bacteria (SRB) Trends Over Time

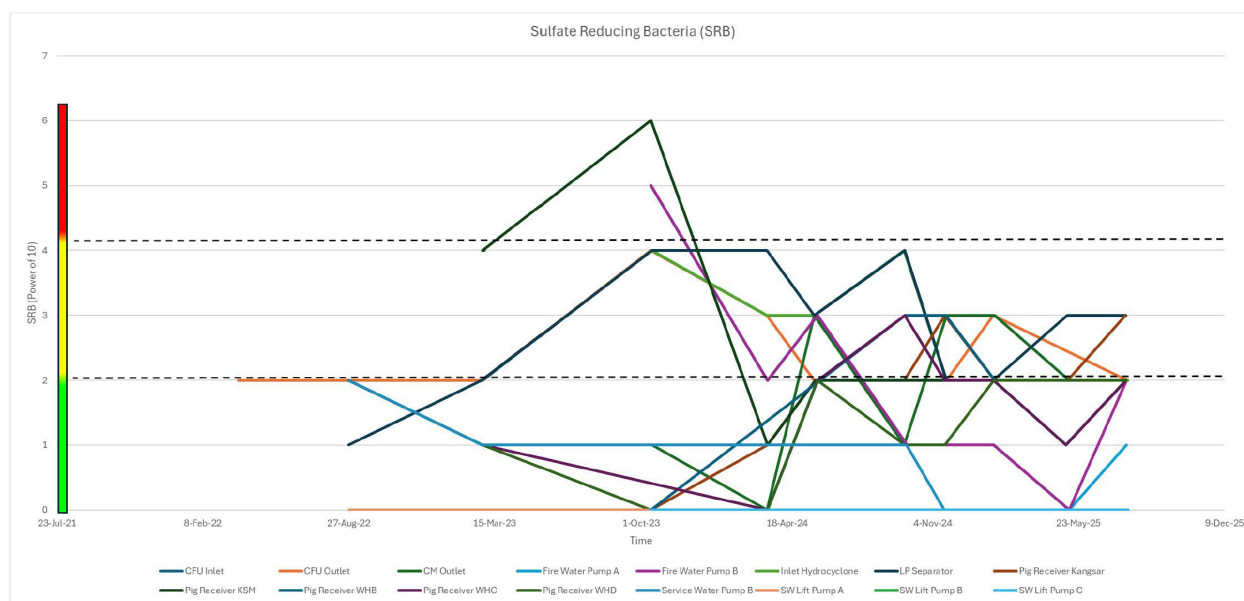


Figure 1—SRB trends over time for sampling points at produced water and seawater treatment systems

qPCR data for sulfate reducing bacteria were tracked from early 2022 through mid-2025 across 15 sampling points within the seawater treatment and produced water handling system (Figure 1).

At the initiation of monitoring (Q1-Q2 2022), SRB numbers showed considerable variability, with several locations exceeding the upper threshold of 10^5 cells/mL (log scale ~5-6), indicative of active and established SRB communities. This variability likely reflected the absence of systematic biocide control measures and the presence of pre-existing biofilms in stagnant or low-flow zones such as pig receivers and filter outlets.

During the first year, intermittent peaks were observed, particularly in samples collected from Free Water Pump B, Service Water Pump B, and the Pig Receiver WHC. These excursions were often associated with operational events (e.g., filter changeouts, flow disruptions).

Between late 2023 and early 2024, a marked shift occurred. Across most sampling locations, SRB levels declined steadily and converged into a narrower envelope predominantly between log 1 and log 3 cells/mL. By late 2024, data variability was significantly reduced, with fewer excursions above log 3. This trend corresponded with the establishment and maturation of the biocide programme and consistent application of THPS and Glutaraldehyde, as well as improved residual biocide tracking, introduced in April 2024.

From late 2024 through mid-2025, the SRB gene counts remained within a relatively stable range, with most points oscillating around the lower threshold (log 1-2). Occasional spikes persisted, primarily in pig receiver samples, which are known accumulation points for debris and microbial regrowth between cleaning cycles.

These trends illustrate several important operational insights:

- **Early Variability:** The high initial variability confirms the presence of mature and spatially diverse SRB populations. This is common in offshore systems that have operated for extended periods without continuous molecular monitoring
- **Progressive Suppression:** The gradual convergence into a stable lower envelope suggests effective suppression of SRB proliferation over time. This result demonstrates that a combination of targeted

biocide treatments and consistent monitoring can reduce microbial risk to manageable levels without continuous high-dose chemical application

- **Data-Driven Optimisation:** The integration of qPCR data into the corrosion management process enabled timely interventions. Spikes in SRB counts were used to trigger targeted dosing, preventing extended periods of elevated microbial risk
- **Residual Risk in Low-Flow Areas:** Persisting excursions in pig receivers and other low-flow components highlight that these areas remain at higher risk for localized MIC. This underscores the necessity of both chemical treatment and physical cleaning strategies to achieve sustained control
- **Reduced Variability as an Indicator of Stability:** The narrowing of SRB counts over time provided confidence in the consistency of control measures. This trend can be viewed as a practical indicator of system stability and treatment efficacy

The consistent decline in SRB qPCR values over the 3-year period aligns with observed reductions in corrosion risk indicators and improved treatment efficiency, as corroborated by field corrosion data and biocide consumption trends.

ATP Trends Over Time

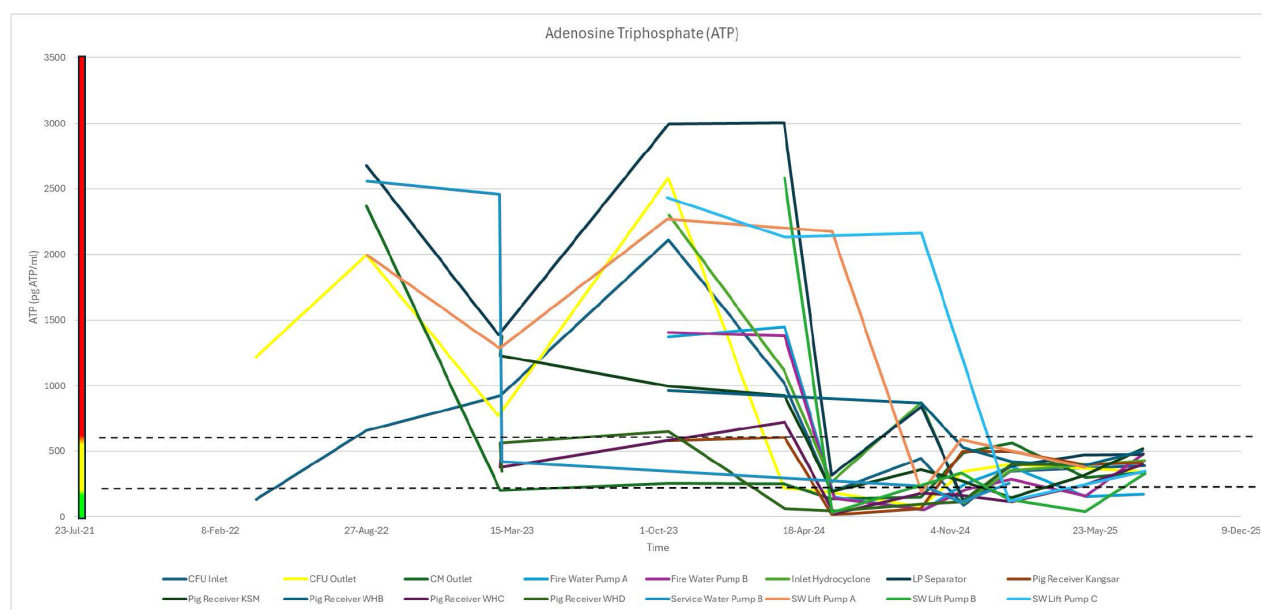


Figure 2—ATP trends over time for sampling points at produced water and seawater treatment systems

The ATP trend data, representing the active microbial biomass present within the system, showed a clear and measurable response to the implementation of the Corrosion Monitoring Programme (CMP). During the initial phase of monitoring, from early 2022 through mid-2023, ATP concentrations were highly variable and frequently exceeded 2,000 pg ATP/mL at several key sampling points. These elevated and fluctuating readings reflected substantial, active microbial populations consistent with the possible presence of mature biofilms and the absence of systematic, continuous biocide control measures during this early period.

Between late 2023 and early 2024, ATP levels exhibited a marked and sustained decline. Most sampling locations converged toward readings below 500 pg ATP/mL. This downward trend closely corresponded with several operational improvements, including the introduction of a structured biocide programme in April 2024, along with the introduction of the tracking of biocide residual concentrations. The rapid and pronounced decline in ATP levels following these interventions underscored the responsiveness of ATP monitoring as a near-real-time indicator of treatment efficacy.

From late 2024 through mid-2025, ATP values remained consistently within a low-activity range. The majority of sampling points recorded ATP concentrations below the 500 pg ATP/mL threshold, indicating effective suppression of active microbial biomass across the seawater and produced water systems. Occasional minor excursions were observed, particularly in pig receiver and low-flow locations, highlighting the inherent challenges of managing microbes in areas prone to stagnant conditions and debris accumulation.

These trends provide several important operational insights:

Rapid Treatment Feedback: ATP monitoring enabled the field operations team to rapidly evaluate the effectiveness of biocide dosing and adjust strategies in near real time, significantly reducing the lag between treatment application and measurable outcomes

Evidence of Disruption: The substantial early reductions in ATP suggest that biocide treatment was effective at disrupting and suppressing planktonic microbial activity, which in this case is used as an indirect indicator of managing MIC risk

Need for Complementary Strategies: Persistent low-level ATP excursions in deadlegs and pig receivers reinforced the importance of coupling chemical treatments with periodic mechanical cleaning or flushing to achieve comprehensive microbial control

Indicator of Long-Term Stability: The convergence and sustained low levels of ATP over time provided a practical and reliable indicator of system stability and the success of the integrated corrosion management programme

Overall, the decline and stabilisation of ATP concentrations demonstrated that the combination of systematic biocide application, residual concentration verification, and proactive microbial monitoring can drive a measurable reduction in active biomass. This capability allows operators to transition from reactive to predictive MIC management and improves confidence in the durability of control measures over extended operational periods.

AMP Trends Over Time

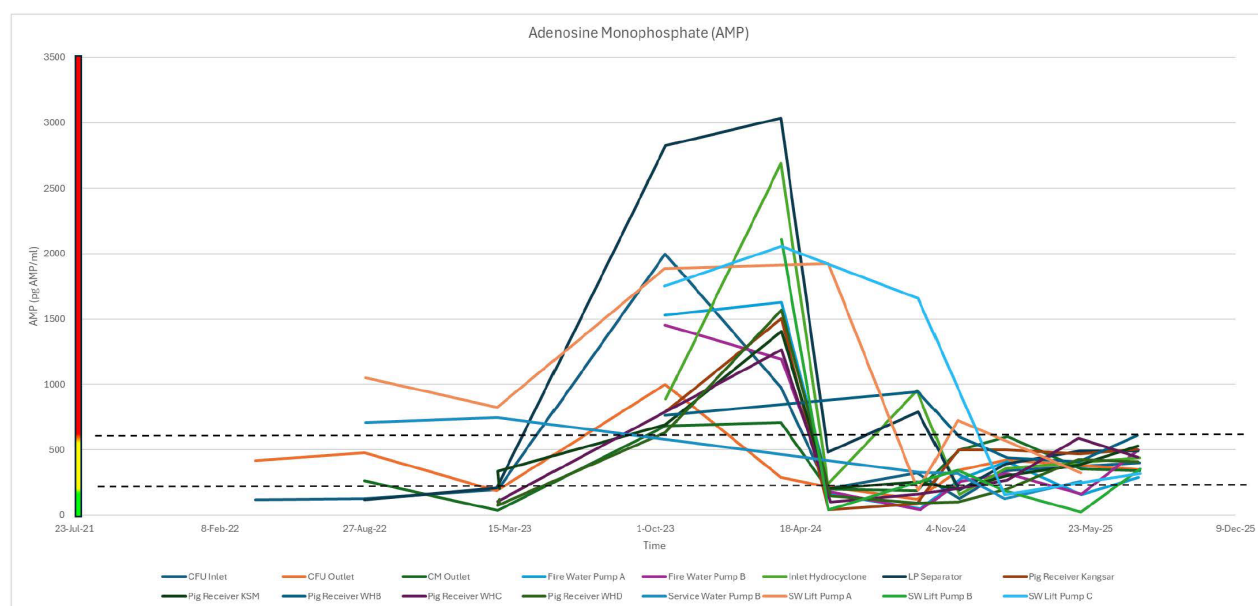


Figure 3—AMP trends over time for sampling points at produced water and seawater treatment systems

The AMP trends, which provide a broader measure of total cellular presence—including stressed, damaged, or non-viable cells—generally followed a similar trajectory to the ATP data over the monitoring period. During the early stages of the programme, AMP concentrations exhibited substantial variability across

sampling points, with several locations consistently exceeding 2,500 pg AMP/mL. These elevated values reflected the presence of a large, established microbial population, encompassing both metabolically active organisms and those persisting in a dormant or inactivated state.

Following the implementation of the systematic biocide rotation programme, AMP readings demonstrated a marked and progressive decline. By late 2023, AMP concentrations had converged into a narrower and substantially lower envelope, indicating that the biocide treatments were effective not only in suppressing active biomass—as reflected by reductions in ATP—but also in significantly reducing the total microbial burden within the system.

This trend is noteworthy because AMP serves as a lagging indicator of historical microbial activity. Elevated AMP can persist even after ATP has been suppressed, due to residual cellular material and incomplete biomass removal. The observed decline therefore suggests that repeated biocide applications, coupled with system flushing and improved biocide residual tracking, contributed to the gradual eradication of accumulated microbial matter over time.

By late 2024, AMP levels at most sampling points had stabilised in a consistently low range, with fewer excursions or episodic spikes. This sustained stability provided a reliable operational marker of treatment efficacy. Importantly, the convergence of AMP trends alongside ATP reductions offered the operator an integrated, multi-dimensional perspective of microbial status—enabling more confident decisions about dosing frequency, concentration, and the timing of intervention.

Furthermore, the dual measurement of ATP and AMP helped distinguish between transient microbial suppression (reflected by rapidly declining ATP) and more durable reductions in total biomass (indicated by the steady contraction of AMP). This distinction proved valuable in understanding the long-term impact of treatment programmes, particularly in zones prone to biofilm formation or intermittent contamination.

Overall, the progressive reduction and stabilisation of AMP levels reinforced the effectiveness of the integrated monitoring and biocide strategy, demonstrating that sustained suppression of both active and residual microbial populations is achievable through data-driven, targeted interventions.

H₂S Trends Over Time

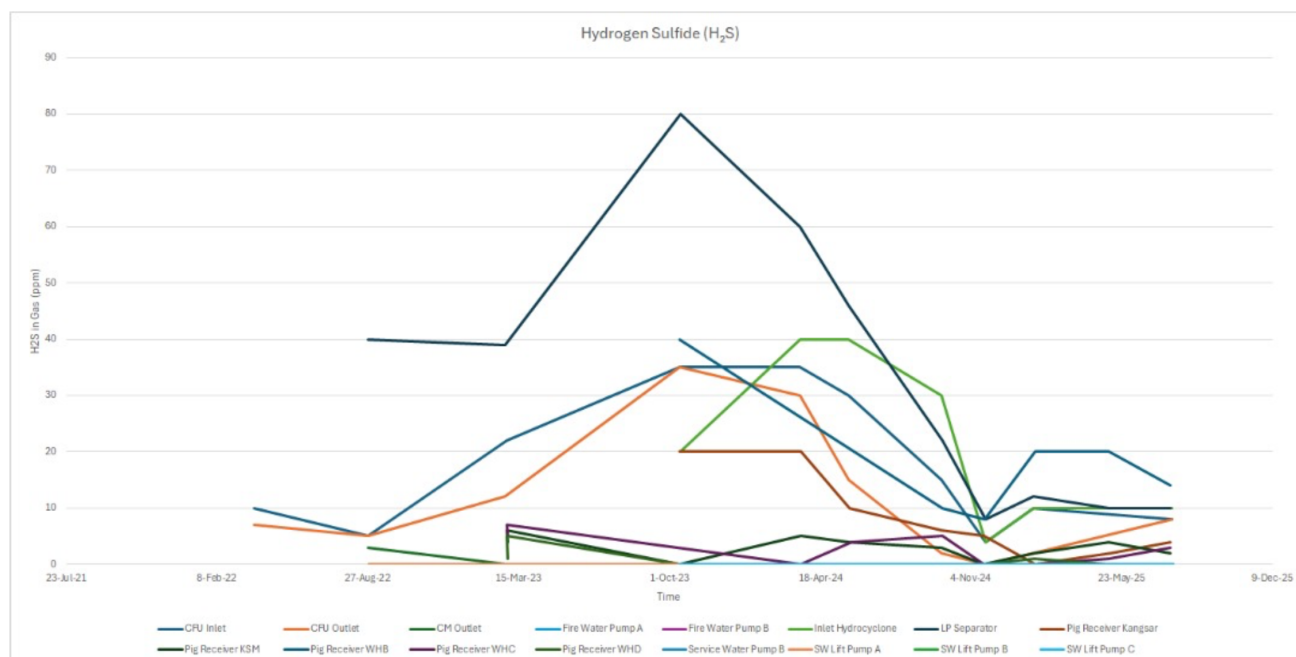


Figure 4—H₂S concentration trends over time for sampling points at produced water and seawater treatment systems

Hydrogen sulfide (H₂S) concentrations were monitored as a direct indicator of sulfate-reducing bacteria metabolic activity and, by extension, as a critical parameter for assessing the risk of MIC. During the initial monitoring period spanning early 2022 to mid-2023, H₂S measurements showed pronounced variability and frequent excursions above operational thresholds. The highest readings were consistently recorded at the Pig Receivers—locations characterised by low-flow regimes and a predisposition to accumulation and stagnation.

Notably, a significant H₂S peak exceeding 70 ppm was observed in late 2023 at the Service Water Pump B. This event coincided temporally with elevated ATP and AMP levels and high SRB numbers. The alignment of these trends reinforced the conclusion that the surge in H₂S generation was microbially driven, validating the need for an integrated and complete monitoring for early detection of MIC threats.

Following the comprehensive rollout of the integrated Corrosion Monitoring Programme (CMP) and the establishment of a systematic biocide rotation strategy in April 2024, H₂S concentrations demonstrated a marked and sustained decline. The majority of sampling points exhibited a rapid reduction in H₂S, with concentrations converging below 10 ppm within weeks of treatment stabilisation. This downward trend mirrored the concurrent suppression of ATP, AMP, and SRB qPCR counts, confirming that microbial populations responsible for H₂S production were effectively targeted and brought under control.

From late 2024 through mid-2025, H₂S levels remained consistently low, showing minimal variability and no significant excursions across the sampling points. The stabilisation of H₂S at a low, predictable baseline provided a clear operational signal that the integrated microbial and chemical treatment strategy was not only effective in reducing active biomass but also in mitigating one of the most aggressive corrosion drivers in offshore production systems.

Several key insights emerge from these trends:

- **Direct Correlation to Microbial Activity:** The temporal alignment between H₂S peaks and elevated microbial markers (ATP, AMP, and SRB) underscores the diagnostic value of using H₂S monitoring as an indicator of sulfate-reducing bacterial resurgence
- **Confirmation of Treatment Efficacy:** The rapid suppression of H₂S levels following the introduction of structured biocide dosing validates the responsiveness of the treatment programme. It also demonstrates that H₂S data can serve as a real-time verification of SRB control measures
- **Indicator of Residual Risk:** While most sampling points stabilised quickly, persistent low-level H₂S detection in pig receivers highlights the importance of ongoing surveillance and the need for physical cleaning interventions in zones prone to SRB entrenchment
- **Evidence of Predictive Control:** The convergence of H₂S, ATP, AMP, and SRB data into consistently low envelopes over time supports the transition from reactive to predictive corrosion management. The ability to anticipate and verify treatment outcomes in near real time marks a significant operational improvement over legacy approaches relying solely on periodic testing

Biocide Residual Efficacy and Tracking

To assess the real-world efficacy and persistence of the biocide treatment protocol, residual biocide concentrations were systematically tracked alongside ATP and SRB measurements over a 72-hour period following each biocide slug. This approach provided critical insight into the duration of antimicrobial activity, the rate of microbial population rebound, and the effectiveness of dosing strategies throughout the produced water treatment system. The Compact Flotation Unit (CFU), located at the tail end of the treatment train, served as a representative location to evaluate system-wide impact.

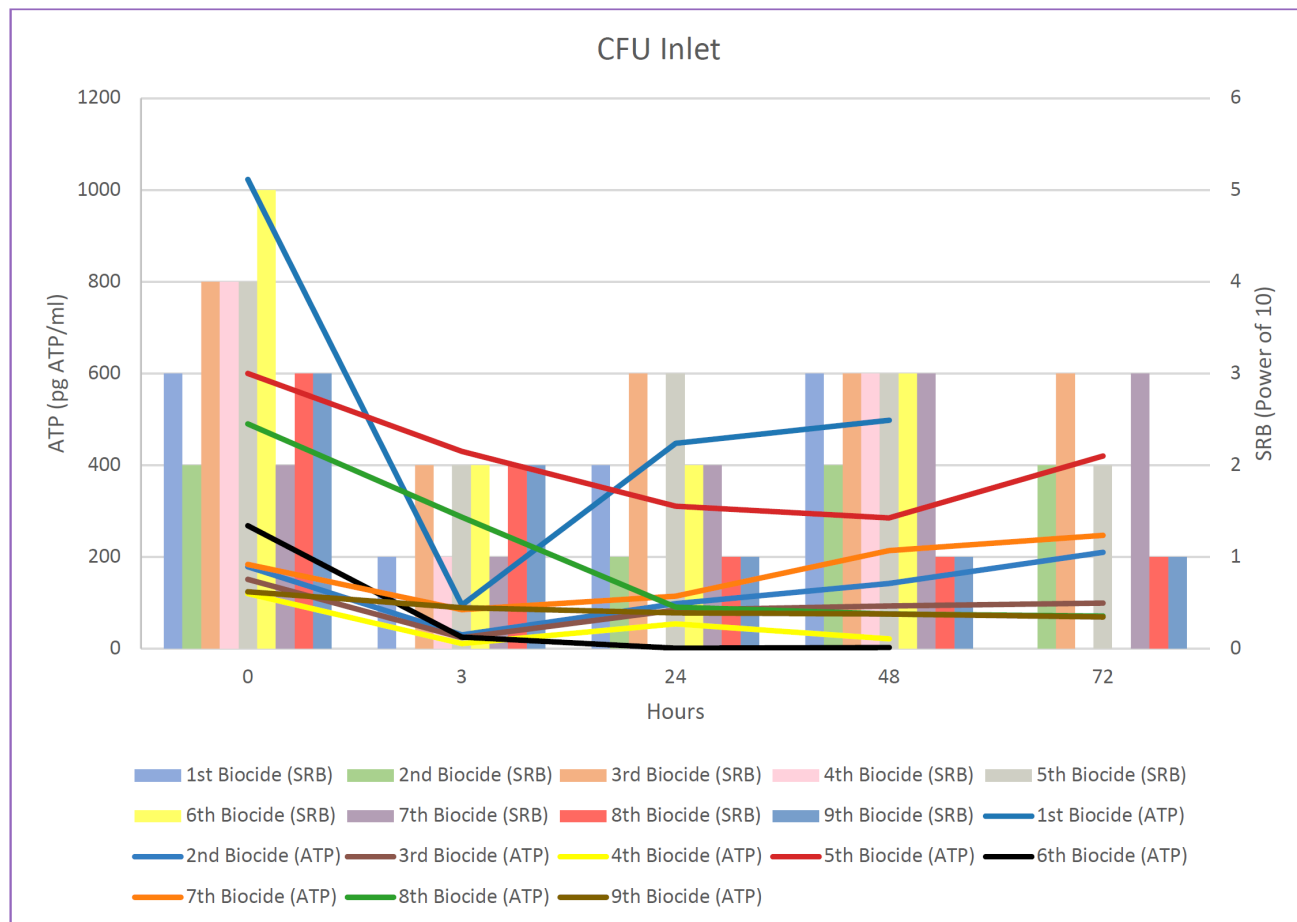
CFU Inlet.

Figure 5—ATP and SRB results at CFU inlet over the duration of 0-72 hours for 9 biocide programs

At the CFU Inlet, the data shows the results of nine biocide programs, each tracked over a 72-hour period. In each program, a significant reduction in both ATP and SRB levels is observed almost immediately after biocide dosing. The most dramatic initial drop is seen after the first biocide slug, where ATP levels fall from approximately 600 pg ATP/mL to below 200 pg ATP/mL within 3 hours, and SRB counts also drop. This rapid response demonstrates the efficacy of the biocide in suppressing the planktonic microbial population.

Over the 72-hour period, a pattern of sustained suppression followed by gradual recovery is evident across all nine programs. ATP and SRB levels remain suppressed for the first 24 to 48 hours before showing a minor upward trend by the 72-hour mark. This trend validates the biocide's ability to provide sustained microbial control and also indicates the appropriate re-dosing window to prevent microbial resurgence. The data from the nine programs collectively provided the operator with a high level of understanding in the treatment protocol's ability to manage microbial activity under varying operational conditions.

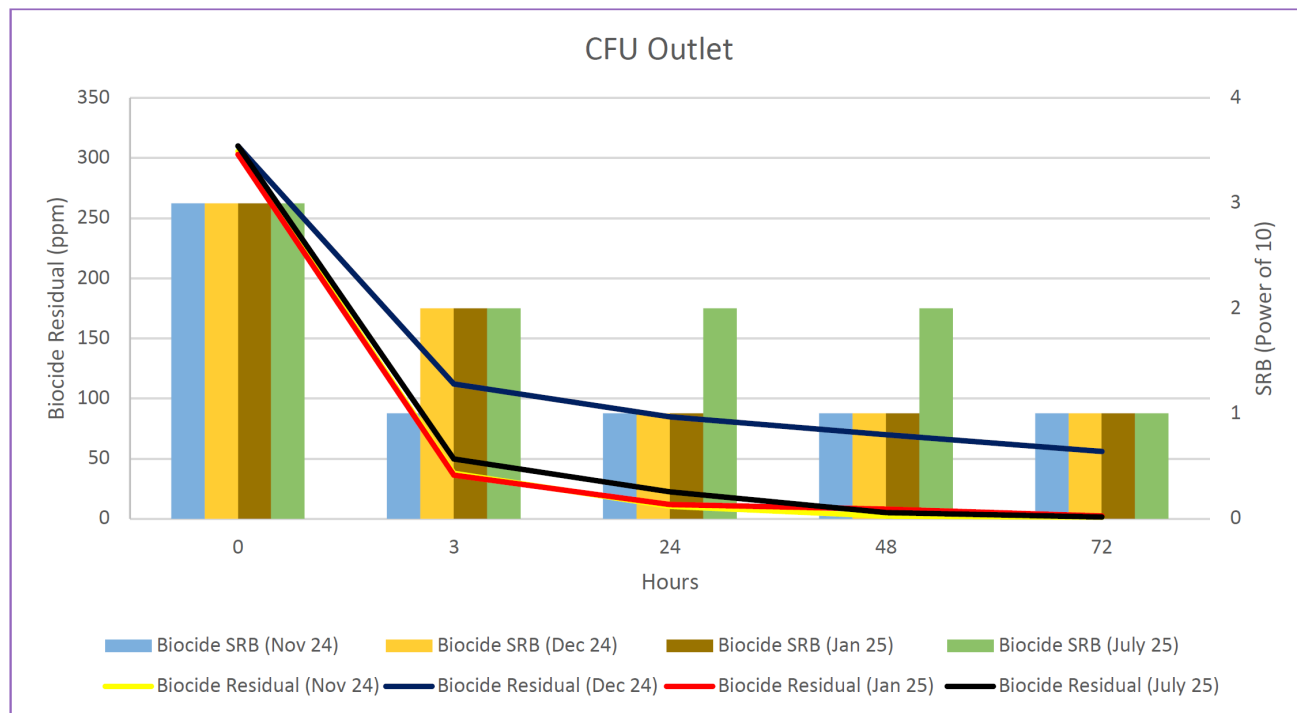
CFU Outlet.

Figure 6—SRB results against biocide residual concentration over the period of 0-72 hours for biocide programs

The CFU outlet data, which tracks biocide residual concentrations alongside corresponding SRB counts, confirms the biocide's ability to propagate downstream and effectively suppress microbial activity. The provided graph shows four distinct biocide programs (November 2024, December 2024, January 2025, and July 2025), each demonstrating a clear inverse relationship between biocide residual concentration and SRB counts.

In each program, a high initial biocide residual level (at 0 hours) corresponds with a low SRB count. As the biocide residual concentration gradually declines over the 72-hour period, a minor increase in SRB count is observed. This trend confirms that the biocide is effective in controlling the SRB population and that the concentration of the biocide is the key factor in its efficacy. The sustained suppression of SRB for the duration of the biocide program demonstrates that the biocide successfully maintained its effectiveness even at a point far from the injection point. This provides strong evidence that the treatment protocol is capable of achieving system-wide microbial control and that the biocide residual tracking is a reliable method for monitoring and optimizing the dosing frequency to prevent microbial regrowth.

Several important conclusions can be drawn from these observations:

- **Validation of Dosing Strategy:** The coordinated tracking of biocide residuals and microbial responses confirmed that the dosing regime achieved both rapid microbial suppression and sustained control over multiple days. This data-driven confirmation is essential for optimising biocide use and avoiding overtreatment or unnecessary chemical expenditure
- **System-Wide Coverage:** The successful suppression of SRB at the CFU Outlet established that biocide slugs were effectively transported throughout the produced water system, overcoming the challenges of dispersion and consumption often encountered in complex offshore processing trains
- **Treatment Predictability:** The consistent trends across multiple dosing events demonstrated that the treatment protocol produced repeatable and predictable outcomes, enabling operators to plan interventions with greater confidence

- **Integration with Molecular Monitoring:** Coupling residual tracking with high-resolution molecular diagnostics provided a robust, multi-layered verification of treatment performance
- **Sustainability of Control:** The fact that microbial populations did not rebound to pre-treatment levels within the monitoring window strongly suggests that the combination of chemical and operational controls created a sustained, hostile environment for microbial re-growth

Water Production and SRB Counts over Time

The interaction between process dynamics and microbial count was evaluated at the inlet of the hydrocyclone by examining the relationship between water production rates and planktonic SRB counts over the course of monthly biocide injection campaigns. The hydrocyclone, as a primary separation point for water from hydrocarbons, represents a critical control point for microbial monitoring due to its role in concentrating water-phase microorganisms and its influence on downstream treatment requirements.

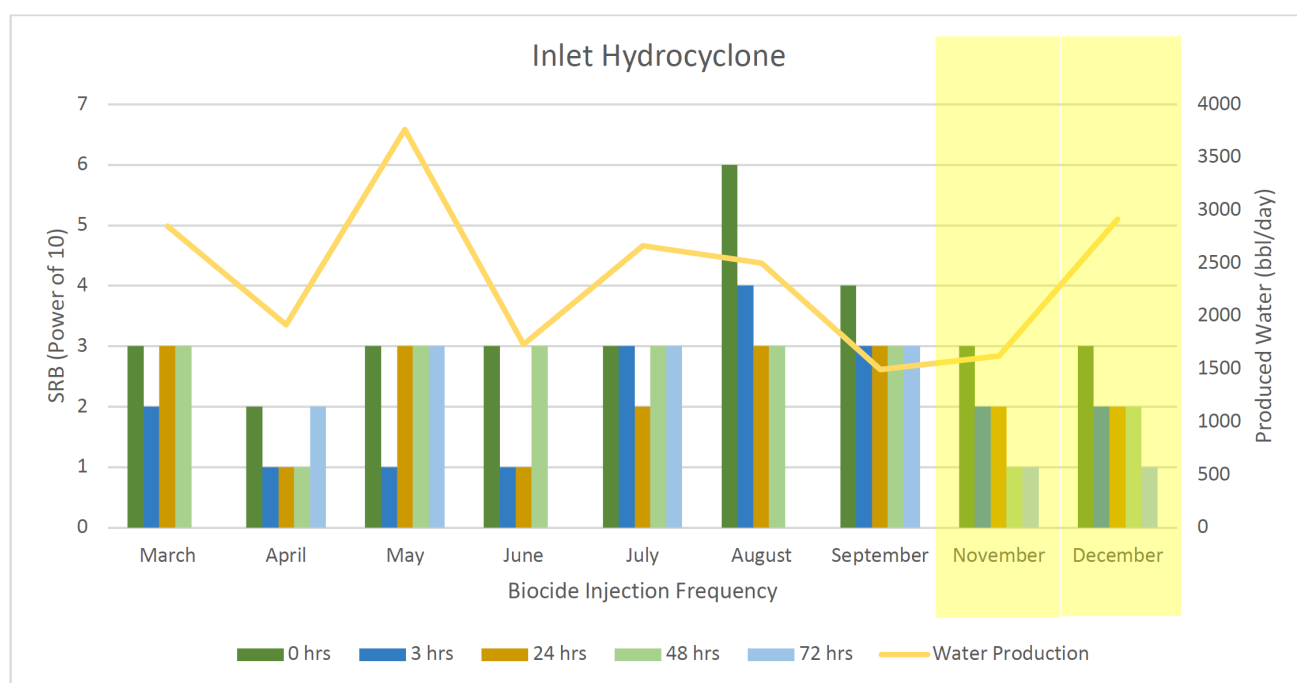


Figure 7—SRB results over varying water production concentration during biocide injection tracked in the period of 0-72 hours

Microbial data were collected at defined intervals—0, 3, 24, 48, and 72 hours after biocide injection and used to assess both immediate and sustained biocide performance under varying production loads. A pronounced spike in water production and SRB counts were recorded in May, the highest water production rate observed in the monitoring period. SRB levels at the 0-hour post-injection point were notably elevated, and while reductions were observed within 3 to 24 hours, counts remained above baseline levels throughout the 72-hour monitoring window. This suggests that the increased water throughput either introduced a higher microbial load from upstream sources or the biocide's initial concentration and contact time were insufficient to achieve full microbial suppression.

In response to the May spike, the biocide injection program was adjusted to adapt and manage the injection efficacy. This adaptation is reflected in the July, August, and September datasets with varying degrees of success in SRB suppression under fluctuating water production rates.

In November, biocide residual monitoring was introduced, highlighted in yellow. The introduction of biocide residual monitoring in November marked a significant improvement in the biocide optimization program. As shown in the provided graph, the data from November and December reveals a different

dynamic compared to the earlier months. The biocide injection dosage was now adjusted based on real-time residual concentrations. This shift in strategy resulted in a more targeted and effective treatment.

The graph for November and December shows that despite a general increase in produced water volume, the SRB counts were maintained at a consistently low level. This indicates that the new approach allowed the operator to ensure that biocide levels were sufficient to suppress bacterial growth, even during periods of high flow and potential increased microbial influx. This proactive approach to biocide dosing, informed by both microbial counts and biocide residuals, proved to be highly effective in maintaining system stability. The success of this strategy demonstrates the value of integrating multiple data streams—microbial activity, process parameters, and chemical residuals—to achieve sustained and optimized MIC control.

The relationship between production rate and SRB load underscores the need to view MIC management not solely as a microbiological challenge but as a process-driven issue as well. Higher water rates can create shorter biocide contact times and potentially higher nutrient loading, all of which favour microbial persistence. The ability to anticipate microbial surges by monitoring process parameters such as water production allows operators to shift from reactive to proactive intervention. This approach minimises the lag between a process change and a treatment adjustment, reducing the likelihood of prolonged microbial issues.

Conclusion and Way Forward

This field case study demonstrates the successful implementation of an integrated, data-driven approach to MIC (Microbiologically Influenced Corrosion) management in offshore produced water and seawater treatment systems. The use of rapid microbial method and molecular microbiology tools—namely ATP, AMP, and qPCR—enabled rapid, accurate profiling of microbial populations, while complementary water chemistry and process data provided critical context for understanding corrosion risk. The use of onsite biocide residual analysis further improved the execution of mitigation strategies.

Over a three-year period (2022-2025), the application of this integrated Corrosion Monitoring Programme (CMP) led to measurable improvements in microbial control and corrosion risk reduction. Key outcomes included:

- A sustained decline in ATP and AMP values, indicating suppression of both active and total microbial biomass
- Significant reductions in SRB counts via qPCR, validating the efficacy of the biocide strategy in targeting corrosion-relevant microbial communities
- Declining and stabilised H₂S levels, confirming a reduction in SRB metabolic activity and MIC drivers
- Consistent microbial suppression across system endpoints, as evidenced by biocide residual tracking at the CFU Inlet and Outlet

The transition from reactive to predictive MIC management has led to enhanced system reliability, more efficient chemical use, and improved asset integrity. These outcomes were enabled not only by the application of modern microbiological tools, but also by strong operational discipline in executing biocide treatments, collecting field data, and adjusting strategies based on real-time insights.

To further enhance MIC management in similar offshore or onshore systems, the following recommendations are proposed:

Implement Slipstream Sessile Sampling for MIC-Relevant Measurement

While planktonic sampling provides valuable insights—particularly in early-stage monitoring—sessile bacteria are more directly responsible for MIC due to their proximity to metal surfaces and their role in biofilm formation. It is recommended that slipstream lines equipped with bio-studs, or corrosion coupons be installed at representative system locations. These units allow the capture and analysis of surface-

attached microbial populations using qPCR and ATP, offering a more direct correlation to corrosion rates and enhancing root cause understanding of MIC events

Deploy Onsite Biocide Selection and Optimisation Units

The selection of the appropriate biocide and its effective dose is critical to long-term microbial control and chemical efficiency. It is recommended that onsite biocide evaluation units using slipstream flow cells be used in conjunction with ATP and SRB qPCR monitoring to simulate field conditions. These units enable field teams to trial various biocide chemistries and concentrations under real operational scenarios, optimising performance before full-scale application. This capability enhances the adaptability and cost-efficiency of MIC programmes

Explore the Use of Longer-Lasting or More Persistent Biocides

Now that biocide residuals are being effectively tracked, operators are in a position to evaluate and adopt biocides with extended half-life and prolonged activity, especially for systems with long residence times or difficult-to-reach zones. Longer-lasting biocides may offer better protection between slug intervals, particularly in low-flow areas where regrowth risk remains high. This could reduce treatment frequency and improve sustainability

Maintain a Triangulated Monitoring Approach

The integration of molecular microbiology, water chemistry, and process data has proven essential to effective MIC management. Each data type contributes a unique dimension:

- Microbiology identifies the biological source of corrosion risk.
- Chemistry reveals the environmental and nutrient conditions that support microbial growth.
- Process data provides context for hydraulic or operational factors that may influence biofilm development.

It is recommended that all three be maintained as part of routine corrosion monitoring programmes to ensure comprehensive and accurate diagnostics.

Build Institutional Knowledge Through Data Archiving and Feedback Loops

To ensure long-term success, historical microbiological and corrosion data should be centralised and trended over time, enabling comparison across campaigns, locations, and treatment events. Feedback from past performance should inform future biocide schedules and microbial monitoring plans, creating a learning loop that continuously improves programme effectiveness.

In summary, this field case demonstrates that a well-structured and adaptive MIC management strategy—anchored by molecular diagnostics, real-time monitoring, and responsive treatment—can significantly improve corrosion outcomes. The integration of tools such as slipstream sessile sampling and onsite biocide optimisation units will further advance the industry's ability to transition from reactive to predictive corrosion control in complex oilfield environments.